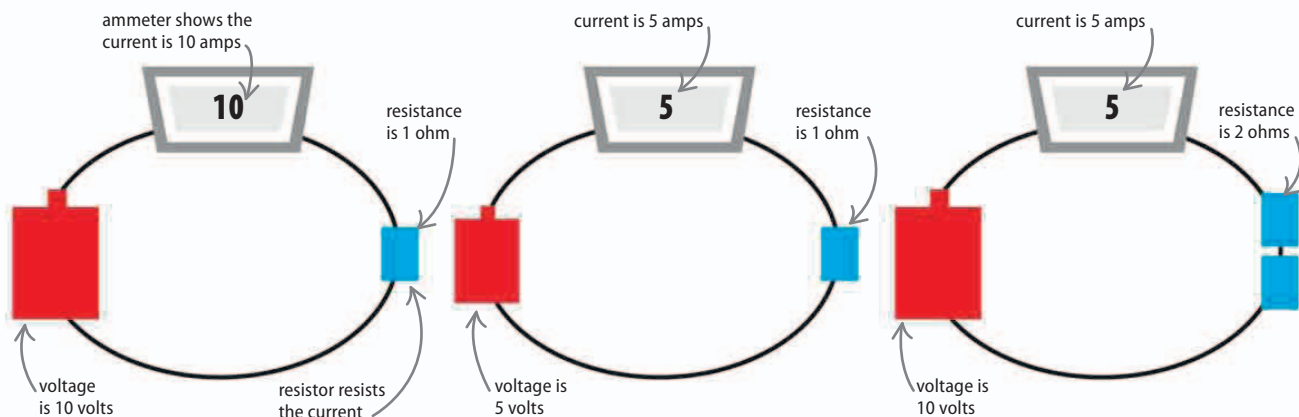


Ohm's Law

Ohm's Law is a formula that shows the relationship between voltage, current, and resistance. Changing the value of one of these three variables will affect the other two. The resistance in a circuit, for example, can be increased by adding an extra component, such as a lamp or a resistor—a device designed to resist the current.

current (amperes; often shortened to "amps" (A)) voltage (volts (V)) resistance (ohms (Ω))

$$\text{Current} = \frac{\text{Voltage}}{\text{Resistance}}$$



△ Circuit 1

In this circuit, the battery provides 10 volts and there is a resistance of just 1 ohm, so the current is 10 amps.

△ Circuit 2

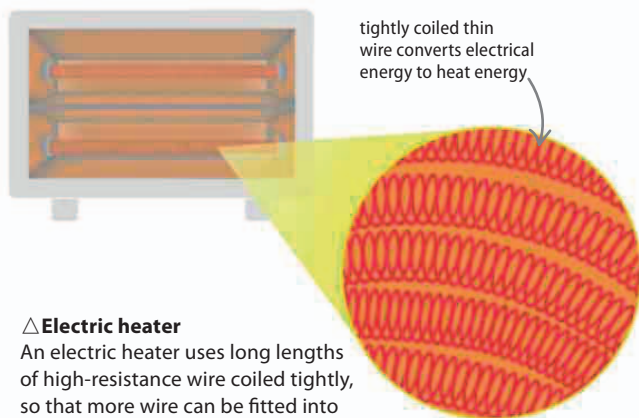
In this second circuit, there is still 1 ohm of resistance, but the voltage has been halved, which reduces the current to 5 amps.

△ Circuit 3

Here, the voltage is again 10 volts, but another 1 ohm resistor has been added, which reduces the current to 5 amps.

Electric heat and light

When electricity flows along a conductor, the resistance that occurs converts some of the electrical energy into heat and sometimes light. The amount of resistance and heat produced can be increased by using a high-resistance wire.



△ Electric heater

An electric heater uses long lengths of high-resistance wire coiled tightly, so that more wire can be fitted into the heater, generating more heat.

REAL WORLD

Superconductors

Certain materials lose practically all of their electrical resistance at very low temperatures. This phenomenon, called superconductivity, can be used to create very efficient electromagnets. These powerful superconducting electromagnets are used in Magnetic Resonance Imaging (MRI) scanners in medicine, in large particle colliders, and in some magnetic levitation (Maglev) rail vehicles, including this Japanese train.

